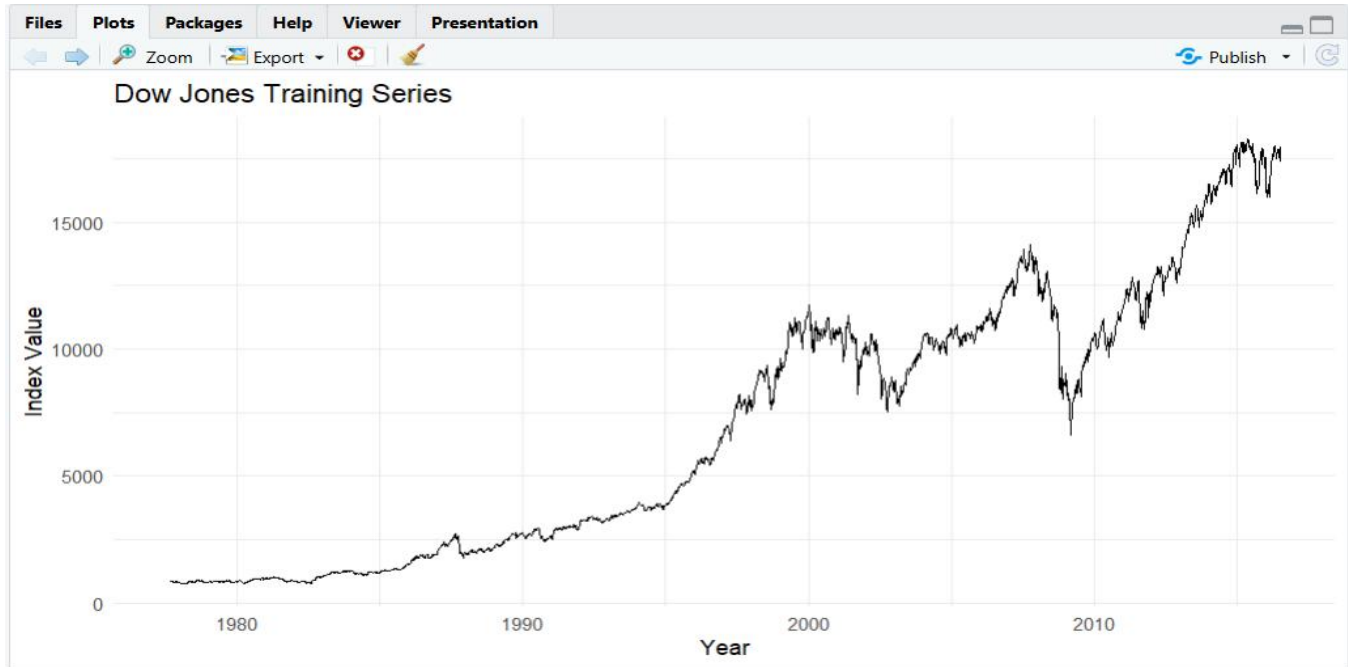


CSCE 5380: DATA MINING

Project 03 – Report: Time Series Analysis

❖ TASK01: Making Time-Series data Stationary:

Given the Dow Jones financial data from 9th February 1977 to 7th January 2016, find the best ARIMA model suited. As the training data is weekly data in the time series conversion we used frequency as 52. The plotted the original time series graph of Dow Jones data:



From the image we can say that the series is not stationary as there is a trend. The price is increasing with time. Let's confirm this using ADF test to prove that series is not stationary, with results as below:

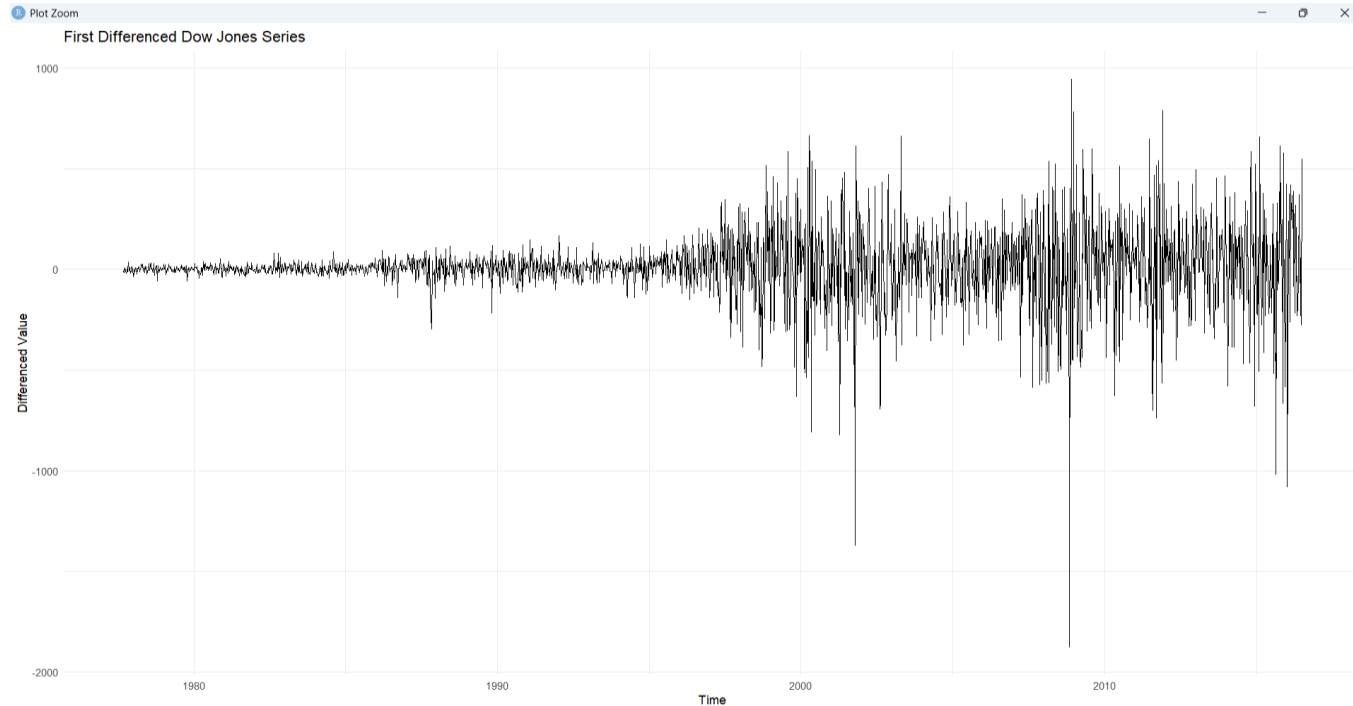
ADF test on original series:

```
> adf_original <- adf.test(dow_ts, k = 12)
> print(adf_original)
```

Augmented Dickey-Fuller Test

```
data: dow_ts
Dickey-Fuller = -2.1164, Lag order = 12, p-value = 0.529
alternative hypothesis: stationary
```

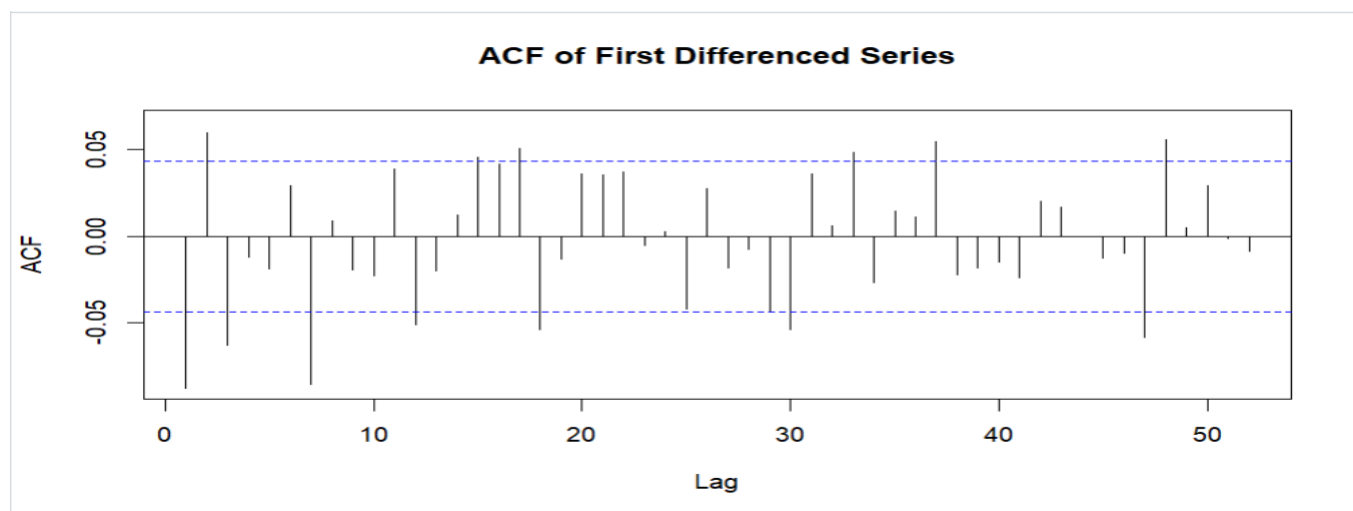
As we can see that the $p > 0.5$, we cannot reject the null hypothesis, which is time series is not stationary. So, we accept it and conclude that series is NOT Stationary.

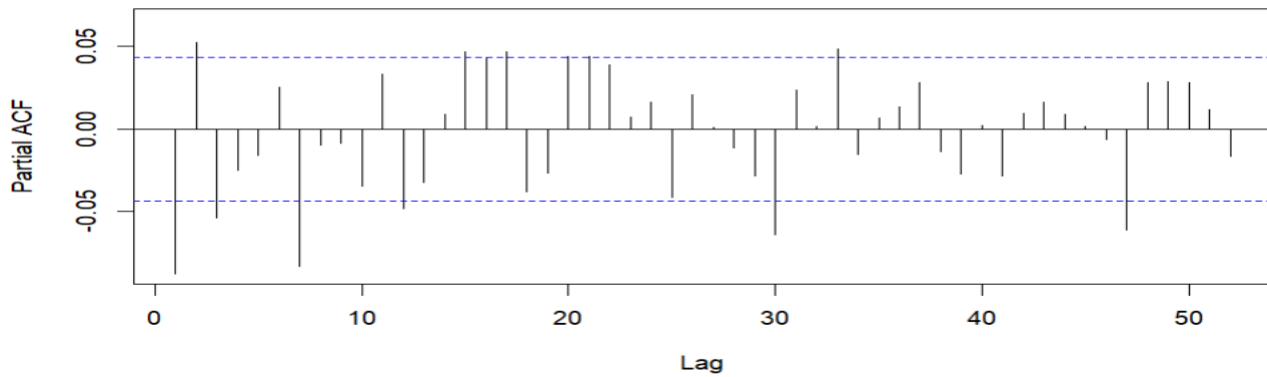
Removing Trend to make the Time series stationary:***i. Now Let's start with the 1st differencing of the time series:***

As we can see from the plot that after the differencing the time series, the mean is constant but the variance is changing at from 2000's, compared before 2000's. So we can conclude that the series is not stationary. Although ADF may give the result as stationary, as it considers unit test instead of the whole time series.

Here The derived ACF and PACF Plots are as follows:

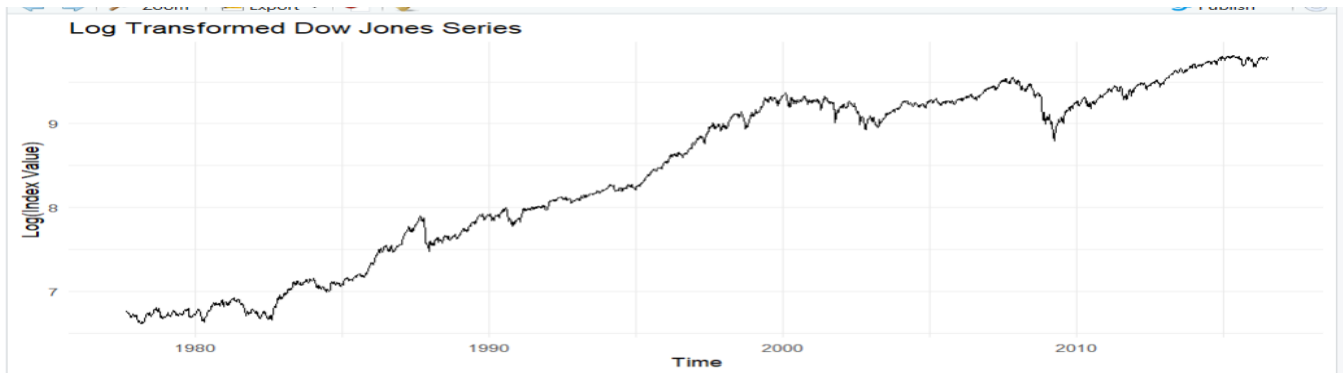
Although the plot doesn't suggest trend which is eliminated due to differencing and seem like stationary. I think log transformation is necessary to deal variance change at the later years, and these also suggest there is mild possibility for seasonality in the data.



PACF of First Differenced Series

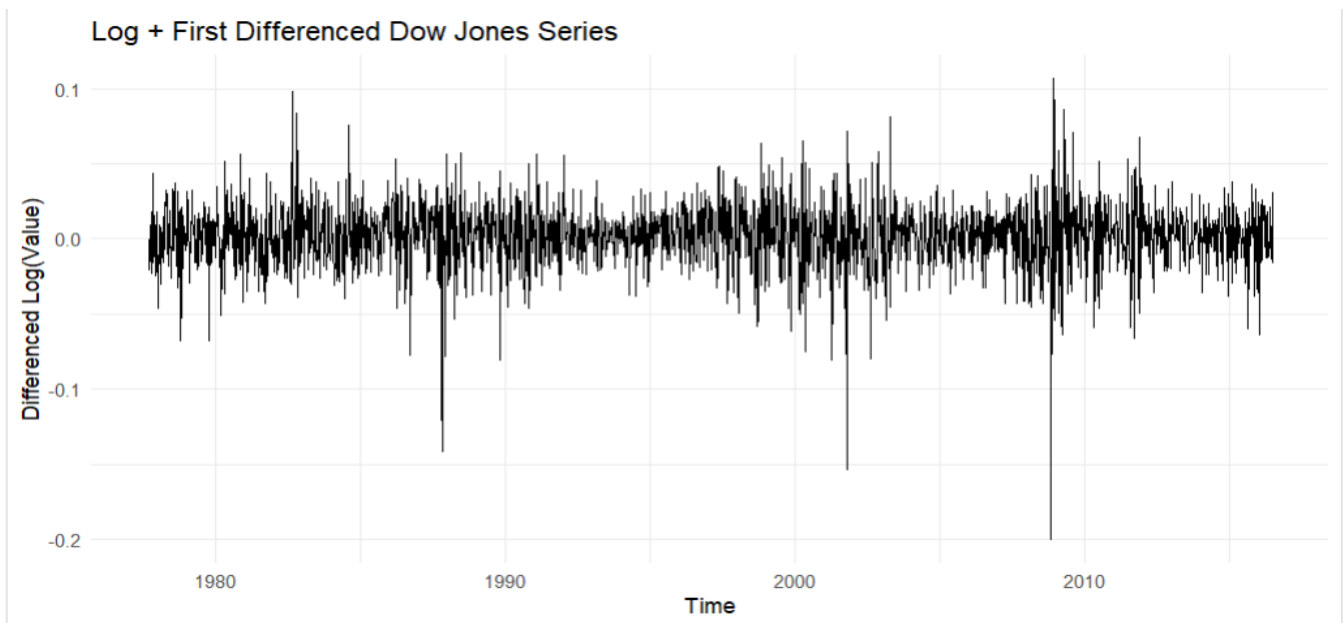
ii. Now Lets do the Log +diff of the time series:

Lets first apply log base e which is also called natural Logarithm to the time series values.



The log series has dealt with the major change in variance in later parts of 2000's and we can see compared to original series and the trend is captured perfectly without losing any information.

Now the differencing of the log series to remove the trend:



As we see in the above differenced Log series the mean and variance both are almost constant which is better than the just the first differencing. Although these are some peak lines which we can consider to be as noise. To be sure now lets confirm the same using Th ADF Test, the results are as follows:

ADF test after log transform + first differencing:

```
> adf_log_diff1 <- adf.test(log_diff1, k = 12)
```

Warning message:

In adf.test(log_diff1, k = 12) : p-value smaller than printed p-value

```
> print(adf_log_diff1)
```

Augmented Dickey-Fuller Test

data: log_diff1

Dickey-Fuller = -13.434, Lag order = 12, p-value = 0.01

alternative hypothesis: stationary

As the $P < 0.5$ which means null hypothesis is not significant, that means we reject the null hypothesis and accept the alternative hypothesis, which means the series is stationary.

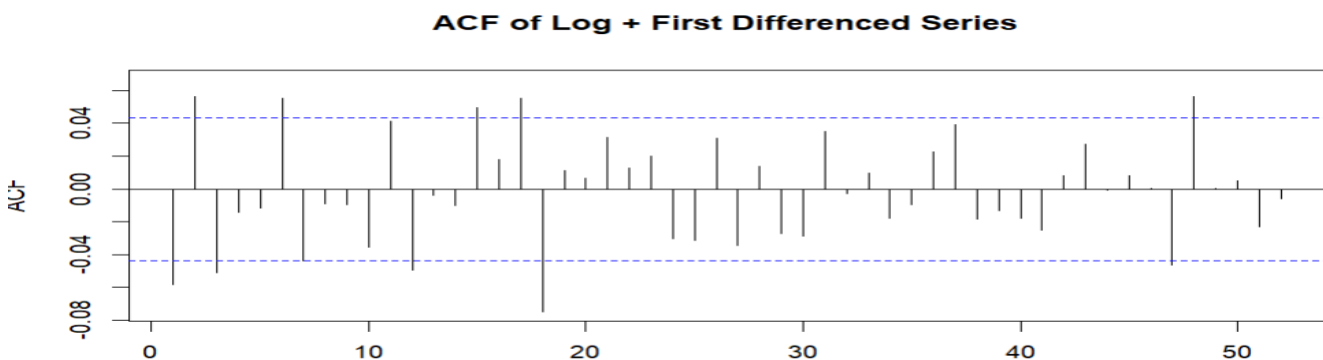
TASK01 Answers:

1. We need one differencing operation to remove trend that is seen thus making series stationary and differenced values are fluctuating around the mean value zero.
2. Yes, we do need the log operation to make variance almost constant after differencing, like compared to just differencing in which we can see the variance is higher after 2000's and vey low before the 2000's
3. I did perform the Dickey Fuller test, ADF on the final satisfied stationary series to confirm whether the series is stationary or not, whose results came to be stationary as you can see in above the TASK01 process.

❖ TASK02: ACF and PACF Plots:

i. ACF Plot:

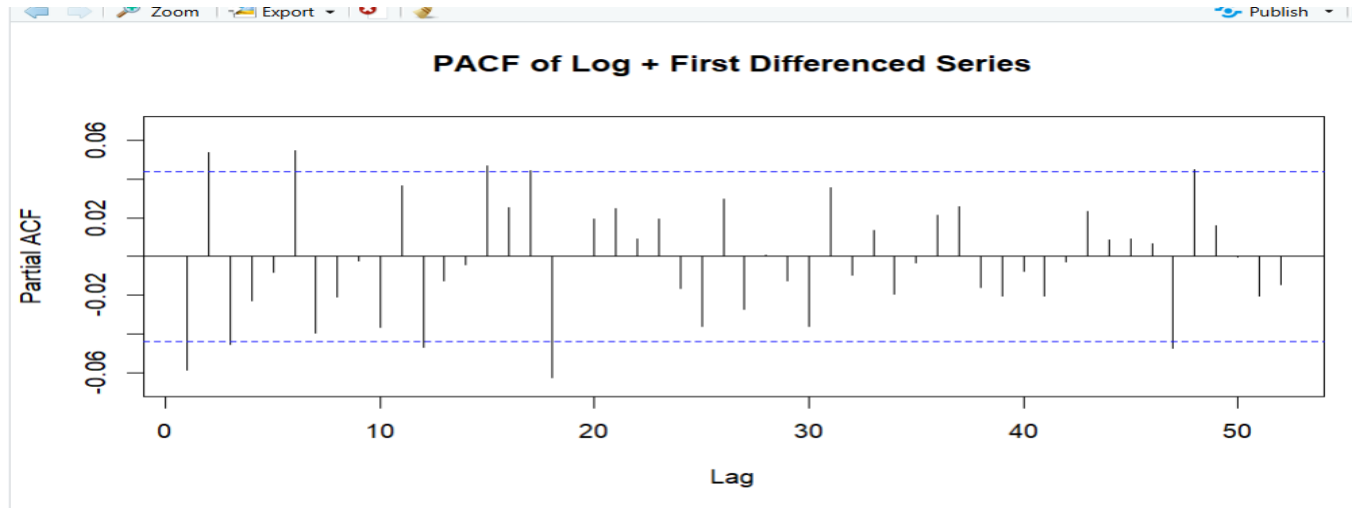
As we concluded that the log + diff series as stationary we now generated the ACF plot for this stationary series.



Now We can see that from ACF Plot lags 1,2,3 are all significant as they cross the blue line and lag 4 is not. So we can see that there is no clear cutoff and as there may other significant spikes which may be white noise or maybe not, after the differencing operation. As this not directly leading to clear q vqlue or MA model we must try all possible values for q starting from 1,2 and finally 3 to get final q value.

ii. PACF Plot:

Same as in case of ACF , we have also generated PACF Plot on the finalized stationary series.



Same as in case of ACF, here too the Lag 1,2,3 are all significant initial lags. So there is no clear cutoff and we have to try all these p values =1,2,3 to get most accurate one.

So we do possibilities of p=1,2,3 and q= 1,2,3, d=1 to get the best Arima, based on parameters like AIC and BIC values. This will lead to 9 possible combinations as below:

Candidate ARIMA models ranked by AIC:

```
> print(results_aic)
```

p	d	q	AIC	BIC
9	3	3	-9608.081	-9563.194
8	3	2	-9604.762	-9565.486
6	2	3	-9603.119	-9563.843
1	1	1	-9602.116	-9579.673
3	1	3	-9600.829	-9567.164
4	2	1	-9600.564	-9572.510
2	1	2	-9600.561	-9572.507
7	3	1	-9600.516	-9566.851
5	2	2	-9598.435	-9564.770

Candidate ARIMA models ranked by BIC:

```
> print(results_bic)
```

p	d	q	AIC	BIC
1	1	1	-9602.116	-9579.673
4	2	1	-9600.564	-9572.510
2	1	2	-9600.561	-9572.507
3	1	3	-9600.829	-9567.164
7	3	1	-9600.516	-9566.851
8	3	2	-9604.762	-9565.486
5	2	2	-9598.435	-9564.770
6	2	3	-9603.119	-9563.843
9	3	3	-9608.081	-9563.194

Now which one to choose ARIMA (3,1,3) or ARIMA (1,1,1). Since we can't decide the values, To select the best p, q I have used another factor named AICc which gives the us the more balanced model. The result of auto Arima came out to be ARIMA (1,1,1), sow I have gone best BIC model.

TASK 02 Answers:

Finally, based on final AICc parameter, like the auto Arima I have decided:

1. $p = 1$, Justification provided above, as there is no clear cut off among the possible values, the model based on AICc parameter is ARIMA (1,1,1)
2. $q = 1$, justification provided above, as there is no clear cut off among the possible values, the model based on AICc parameter is ARIMA (1,1,1).

If we want it to be based on AIC parameter then these values (this is not our case):

1. $p=3$, as best performing model with least AIC value is ARIMA (3,1,3)
2. $q=3$, as best performing model with least AIC value is ARIMA (3,1,3).

Finally, confirming my choice with auto.arima():

Final selected model (based on BIC):

```
> cat("ARIMA(", final_order[1], ",", final_order[2], ",", final_order[3], ")\n", sep = "") > # Confirming with auto.arima
ARIMA(1,1,1) > auto_model <- auto.arima(log_dow)
> print(final_model) >
> print(auto_model)
Series: log_dow
ARIMA(1,1,1) with drift
Series: log_dow
ARIMA(1,1,1) with drift
```

Coefficients:

```
      ar1    ma1  drift
-0.7250  0.6650  0.0015
s.e.  0.0992  0.1072  0.0005
```

```
sigma^2 = 0.0005035: log likelihood = 4805.06
AIC=-9602.12 AICc=-9602.1 BIC=-9579.67
```

Coefficients:

```
      ar1    ma1  drift
-0.7250  0.6650  0.0015
s.e.  0.0992  0.1072  0.0005
```

```
sigma^2 = 0.0005035: log likelihood = 4805.06
AIC=-9602.12 AICc=-9602.1 BIC=-9579.67
```

From the results we can tell the auto. arima() has given ARIMA (1,1,1) as the correct answer.

So, ARIMA (1,1,1) is best, and we should consider $p=1$, $q=1$ as the final answers.

Note: Also, from the residual plots of ACF we can tell that Just the ARIMA is not enough so, our results may not be good enough. We should consider ARIMA+GARCH to getting best results. As GARCH is not covered in our lessons we won't consider that.

❖ TASK03: GENERATE ARIMA MODEL AND TESTING ON TEST DATSET

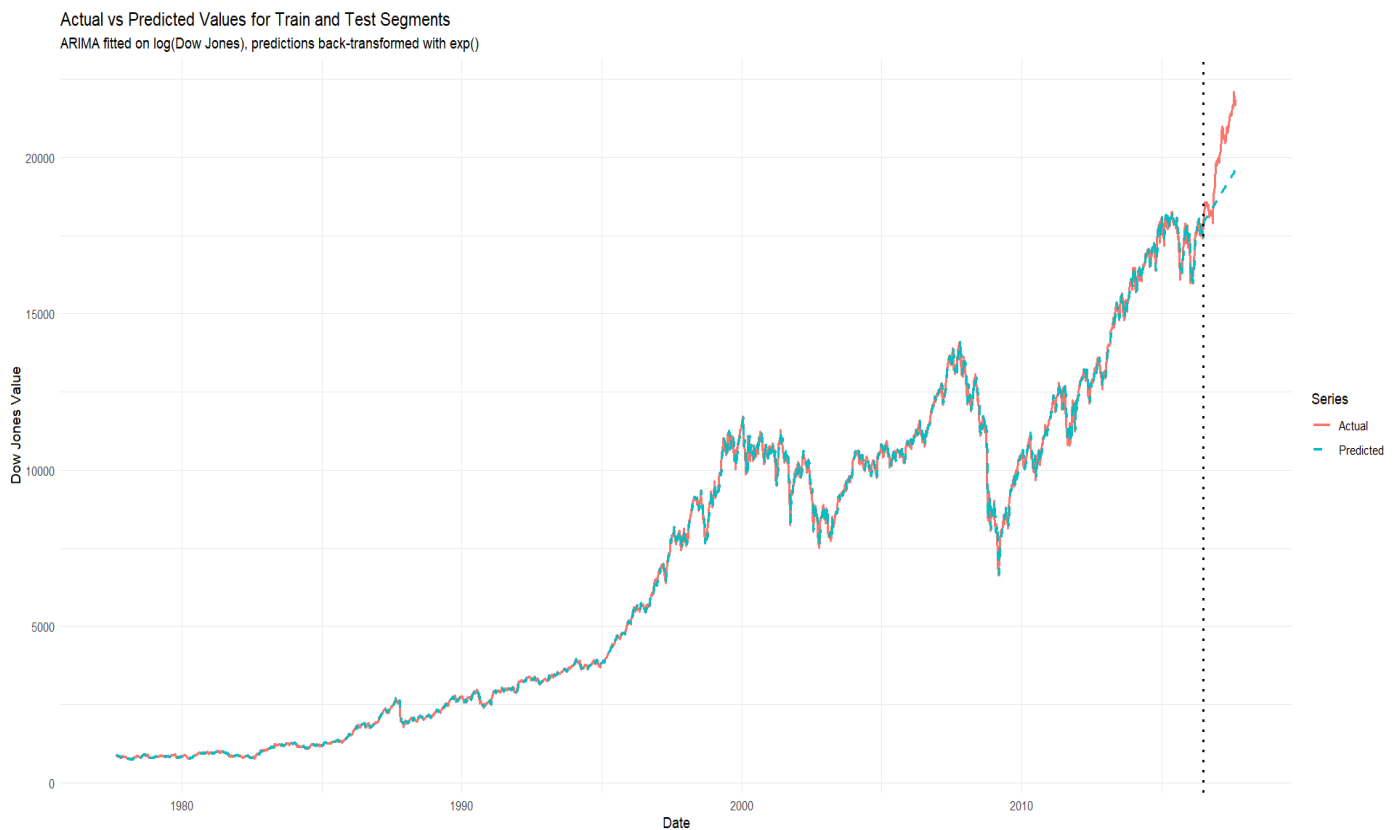
- From the analysis in task02, our final model came out to be ARIMA (1,1,1). Now we load the test set, get its length h, using the forecast of our model to predict the h time stamps, transforming them to original scale and finally measure the rmse, mae and others. The results are as follows:

```
Final Selected Model:
> cat("ARIMA(",
+     final_order[1], ", ",
+     final_order[2], ", ",
+     final_order[3], ")\\n", sep="")
ARIMA(1,1,1)
> cat("\\nTest set accuracy:\\n")

Test set accuracy:
> cat("RMSE :", rmse, "\\n")
RMSE : 1462.662
> cat("MAE :", mae, "\\n")
MAE : 1247.775
> cat("MAPE :", mape, "%\\n")
MAPE : 6.020079 %
```

The Rmse Value came out to be 1462.662

- Now Lets move on the plots of training and testing , we have plot both training and testing in same graph with (actual V/s predicted in testing):



As asked, in the plot blue dotted line is trained on training data and forecasted for the next 30 values. While the red line is the actual training and test data. Now to get even a more clear look at testing :

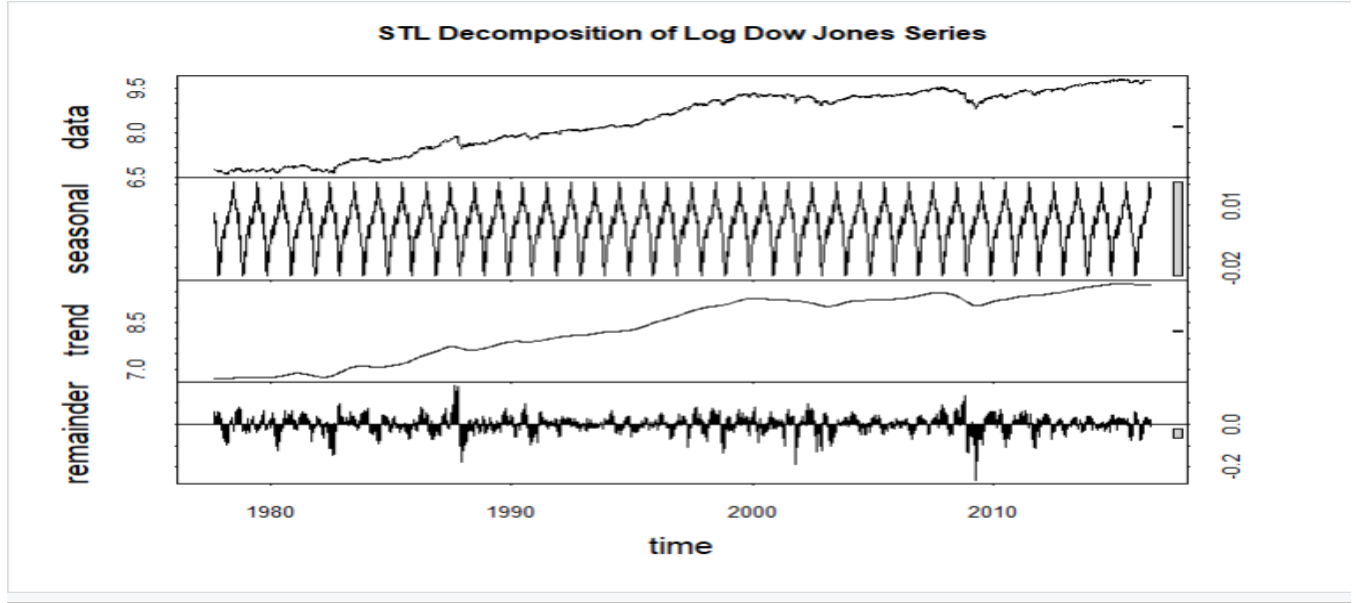
as the error values for ARIMA (3,1,3) are higher than ARIMA (1,1,1) this once again confirms that $p=1$, $q=1$ are correct values and the best model parameters.

❖ TASK04: ACCOUNTING SEASONALITY INTO THE MODEL

1. Extracting the Seasonal Signal:

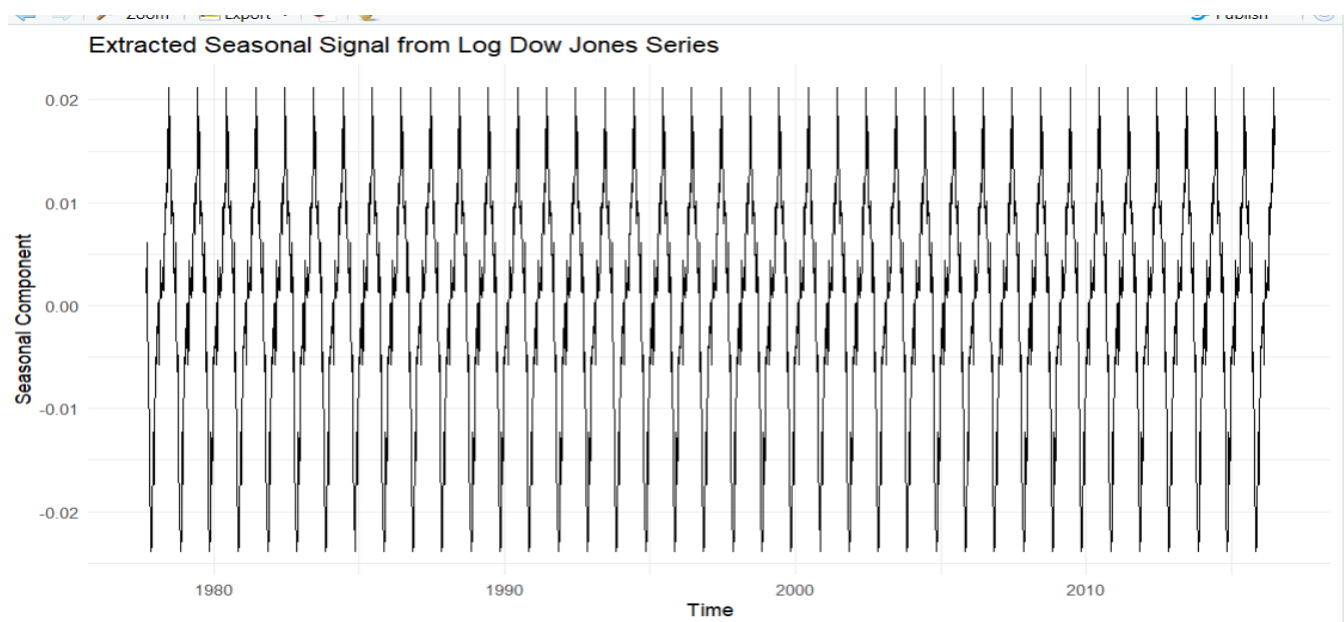
To extract the seasonal signal, I have first decomposed the timeseries using the `decompose()` function and just extracted the seasonal signal. The plots pasted below will give an idea.

The decompose of time series plot:



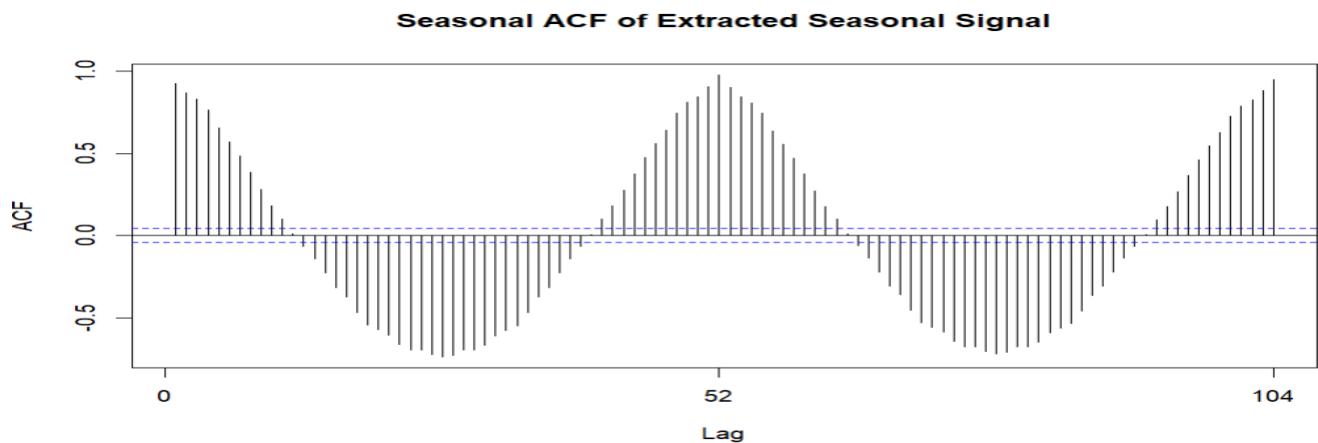
this gives that time series not has trend but also the seasonality so we have to use the sarima instead of regular arima to predicted test values.

Extracted Seasonal signal Plot:



2. ACF and PACF plots on the seasonal signal:

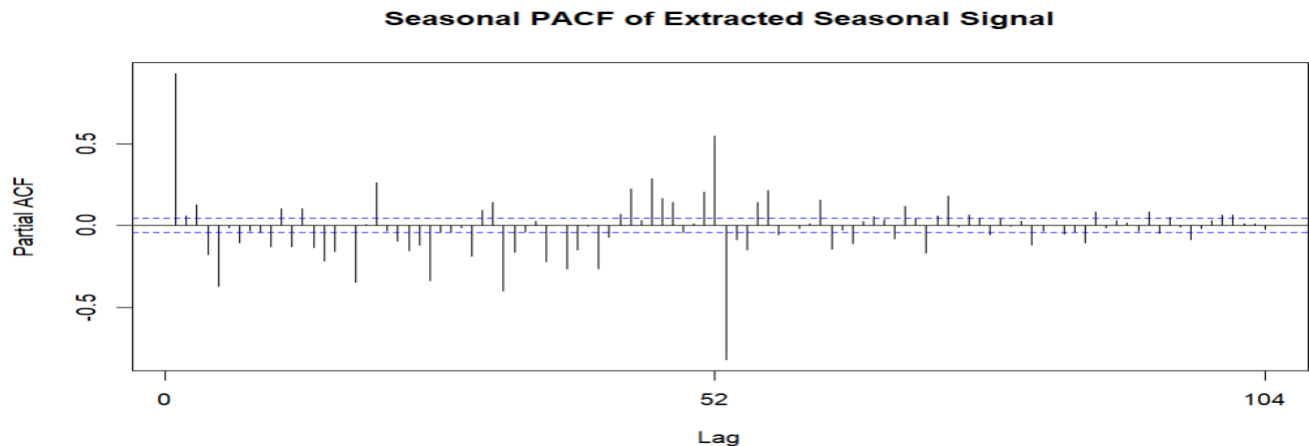
The seasonal signal ACF Plot:



Here $Q=0$

Justification: Here we see it shows very repeated sinusoidal periodic behaviour, since our frequency is 52, it repeats a cycle every 52 lags. It does not cut off at seasonal lags 52, or 104 and flows smoothly in a sine wave manner indicates seasonal persistence or AR type seasonality instead of moving average MA. So, MA is not likely the dominant factor so we can interpret $Q = 0$.

The seasonal signal PACF Plot:



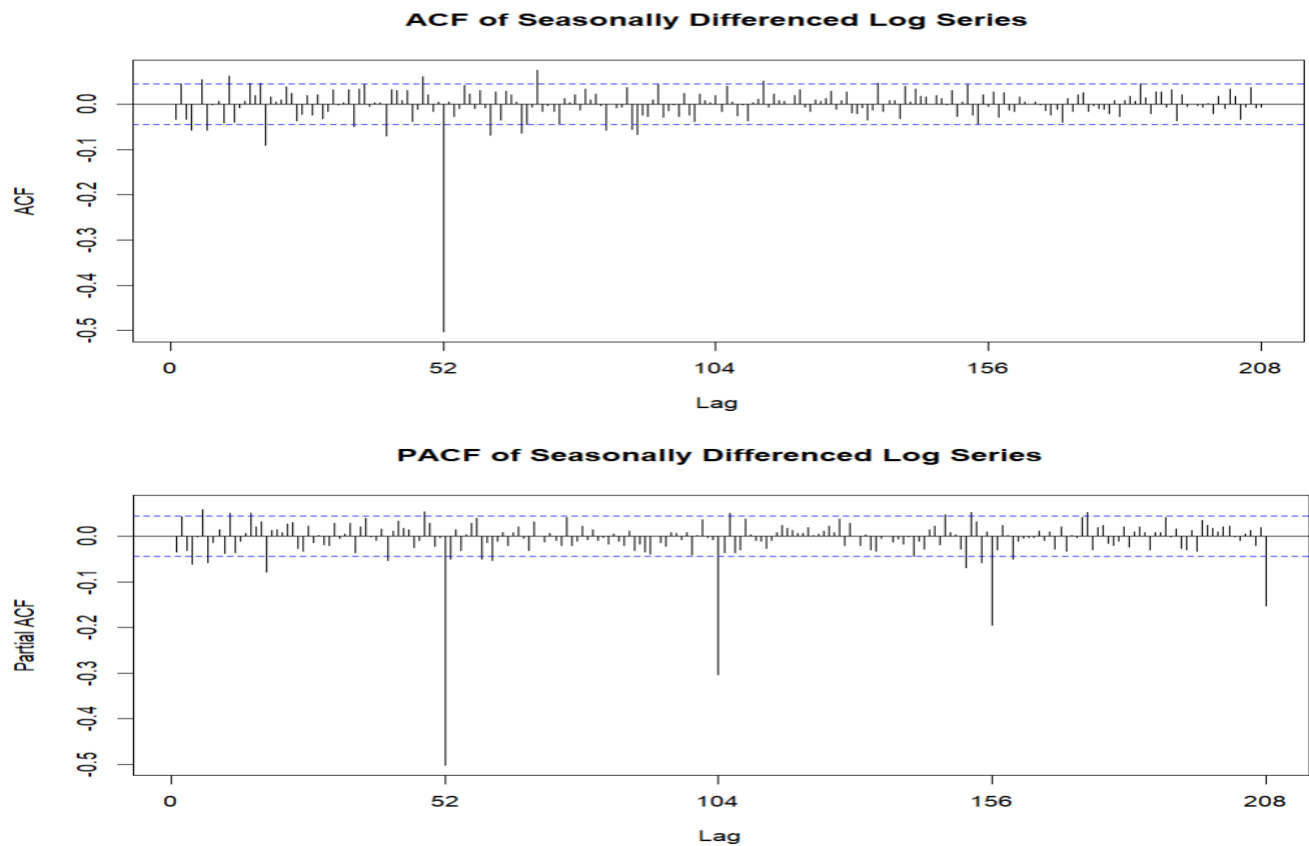
Here we can say that $P=1$

Justification: As we can in the above plot at lags 52, 104 which is our seasonal spikes. There is one significant spike at lag 52, but not at lag 104. So we can say the model is likely of AR (1) for seasonality meaning the value of $P=1$.

As there is a seasonality at lags 52 in series as per plots, which decays very slowly as per ACF, we do $D=1$.

Note /Doubt: These above P, Q are based on the seasonal signal as requested in the document, but not the seasonal differenced original series. From the original differenced of seasonal differenced log

series, its quite contradictory, i.e., $\text{diff}(\text{diff}(\log_dow, \text{lag}=52), \text{differences}=1)$ has given ACF cut off for $Q=1$, and no significant PACF cut off at any of the lags at 52,104, 156,208 all i.e., all lags are significant, so $P=0$. I have asked in teams but got no reply. So I have documented both.



3. Integrating seasonal component and re forecast test values and plotting them:

As we have above values of $P=1$, $D=1$, $Q=0$ we train the $\text{SARIMA}(1,1,1)(1,1,0)[52]$. The training results are as follows:

```
Seasonal ARIMA model summary:
> print(summary(sarima_model))
Series: log_dow
ARIMA(1,1,1)(1,1,0)[52]
```

Coefficients:

	ar1	ma1	sar1
	-0.7200	0.6688	-0.5121
s.e.	0.1262	0.1351	0.0193

```
sigma^2 = 0.0007531: log likelihood = 4278.9
AIC=-8549.8 AICc=-8549.78 BIC=-8527.46
```

Training set error measures:

	ME	RMSE	MAE	MPE	MAPE
Training set	-1.213004e-05	0.02706049	0.02000079	-0.0001095457	0.2389752
	MASE	ACF1			
Training set	0.1453926	0.002791437			

>

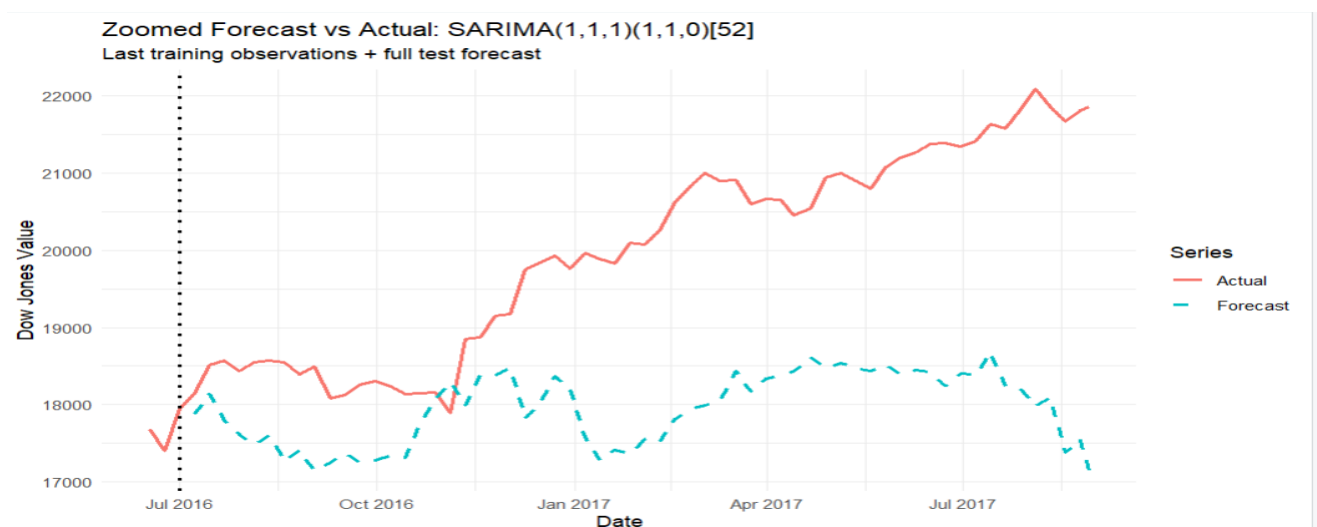
Now similar to task 3, we fore cast log values convert them and plot and against ovginals vales and also get the rmse metrics.

```
>
> cat("\nSeasonal model test accuracy:\n")
Seasonal model test accuracy:
> cat("RMSE :", sarima_rmse, "\n")
RMSE : 2350.997
> cat("MAE :", sarima_mae, "\n")
MAE : 2057.437
> cat("MAPE :", sarima_mape, "%\n")
MAPE : 9.986223 %
> |

> # -----
> # 4) Compare seasonal vs non-seasonal model
> # -----
> cat("\nModel comparison:\n")
Model comparison:
> cat("Non-seasonal ARIMA AIC :", AIC(final_model), "\n")
Non-seasonal ARIMA AIC : -9602.116
> cat("Seasonal ARIMA AIC :", AIC(sarima_model), "\n")
Seasonal ARIMA AIC : -8549.803
> cat("Non-seasonal RMSE :", rmse, "\n")
Non-seasonal RMSE : 1462.662
> cat("Seasonal RMSE :", sarima_rmse, "\n")
Seasonal RMSE : 2350.997
>
```

The rmse with seasonal came out be 2350.997 which has been increased compared to without seasonal, i.e., the performance of the model has been degraded due to adding seasonality to Arima.

The Plots:



As you can see over here dotted line is predicted forecast values, we can see that trend is not captured clearly due to addition of seasonality.

Here I have completely done with all the requirements of the project 3

Extra: As I must consider that differenced of seasonal differenced log series for P,Q value we got $P=0$, and $Q=1$ when rather compared to just the seasonal signal. Now the below results are for $ARIMA(1,1,1)(0,1,1)[52]$ which surprisingly has better results. In code I have set these value for P,Q, instead o the values that are based just solely on the seasonality component.

```
> cat("\nSeasonal model test accuracy:\n")
Seasonal model test accuracy:
> cat("RMSE :", sarima_rmse, "\n")
RMSE : 1644.333
> cat("MAE :", sarima_mae, "\n")
MAE : 1454.605
> cat("MAPE :", sarima_mape, "%\n")
MAPE : 7.059851 %
>

> # -----
> # 4) Compare seasonal vs non-seasonal model
> # -----
> cat("\nModel comparison:\n")
Model comparison:
> cat("Non-seasonal ARIMA AIC :", AIC(final_model), "\n")
Non-seasonal ARIMA AIC : -9602.116
> cat("Seasonal ARIMA AIC :", AIC(sarima_model), "\n")
Seasonal ARIMA AIC : -9181.902
> cat("Non-seasonal RMSE :", rmse, "\n")
Non-seasonal RMSE : 1462.662
> cat("Seasonal RMSE :", sarima_rmse, "\n")
Seasonal RMSE : 1644.333
>
```

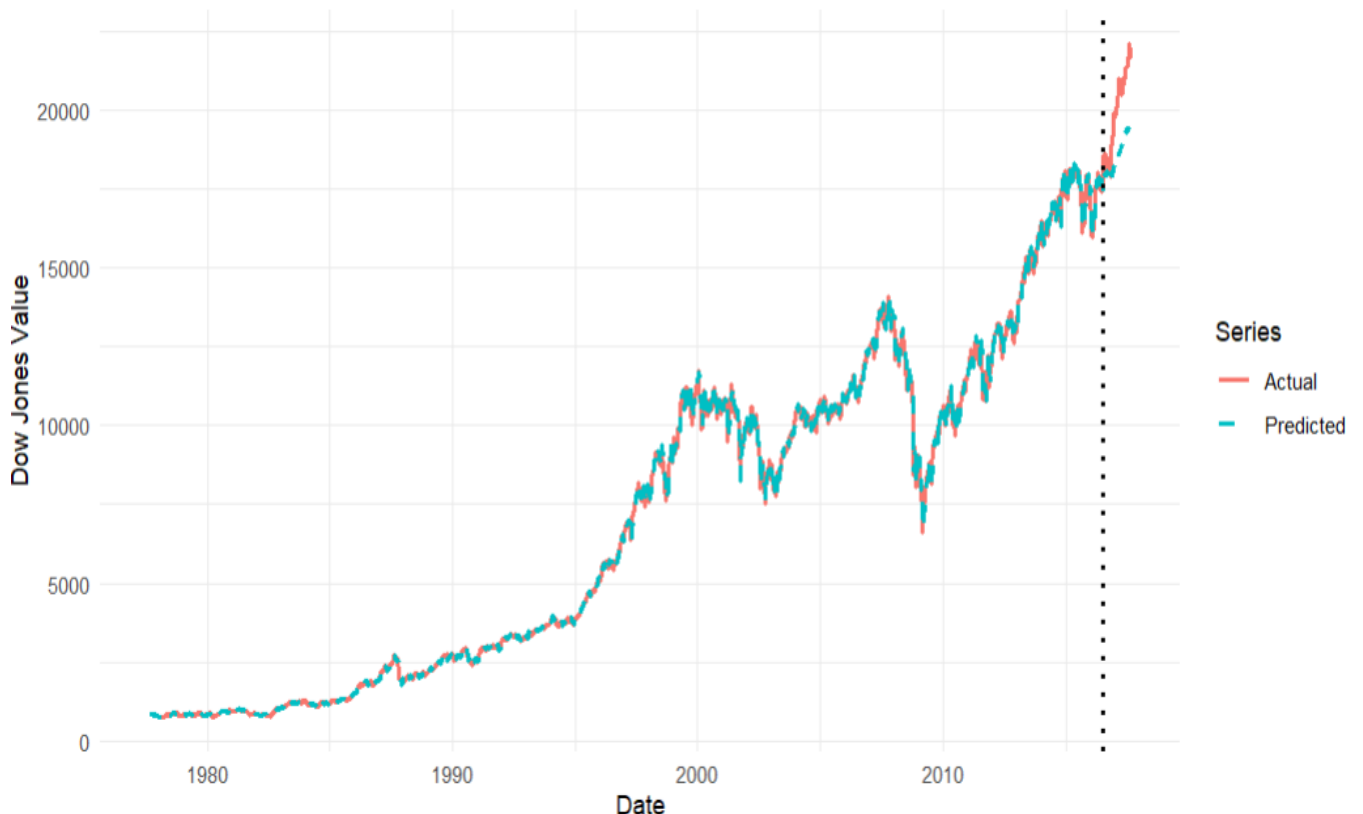
As we can see here rmse for $ARIMA(1,1,1)(0,1,1)[52]$ is much lower than $ARIMA(1,1,1)(1,1,0)[52]$ which is around 2000's so I think while getting P,Q we should not just consider the seasonal signal but the whole series.

Also compared to modelling with seasonality with out seasonality has better performance.

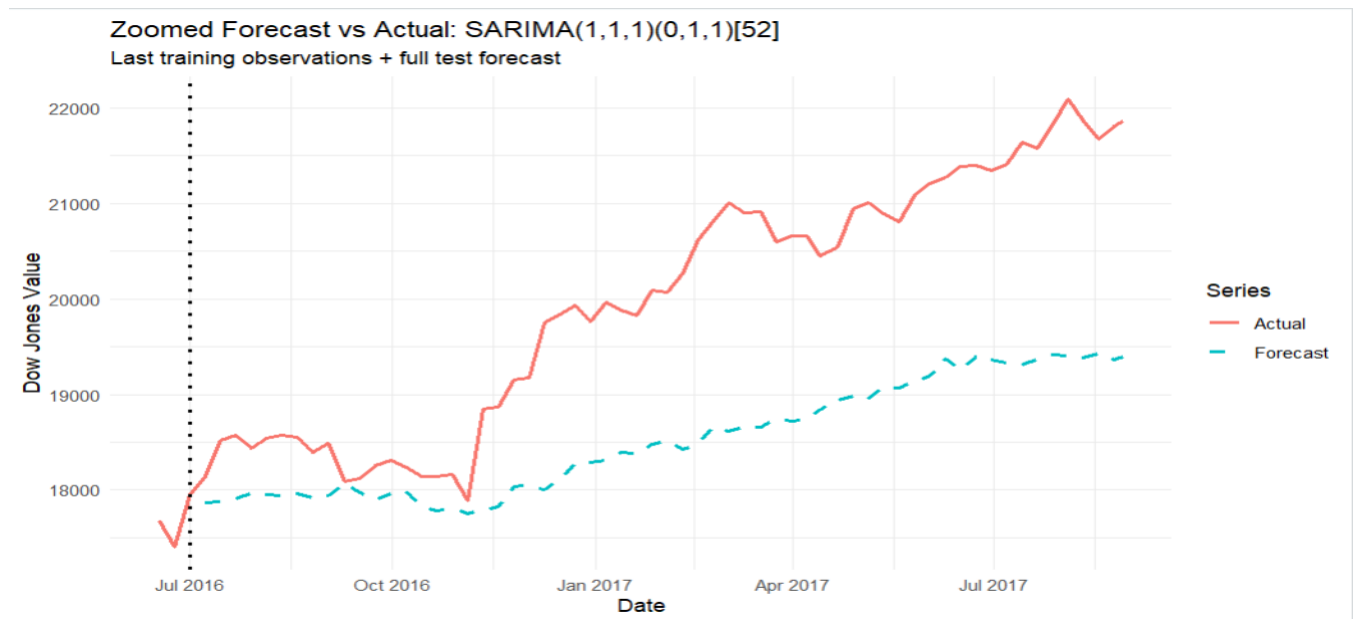
Now lets see the plots:

Seasonal ARIMA: Actual vs Predicted for Train and Test

$SARIMA(1,1,1)(0,1,1)[52]$



The zoomed plot:



From here we can say that the trend is captured successfully, compared to Arima(1,1,1)(1,1,0)

So, I conclude that $P=0$, $Q=1$ for task 4.2 but based on seasonal differenced series, no to go with $P=1$, $Q=0$ based on just seasonal signal.